A DEM Differencing Method for Detecting Geomorphic Changes on Topographically Complex Areas Based on RAV Remote Sensing Techniques

Dou Li, Pengfei Li, Jinfei Hu, Wanqiang Yao, Lu Yan, Hooman Latifi, Bingzhe Tang, and Lifeng Liu

***Abstract*— High-resolution topographic data acquired by remote aerial vehicles (RAVs) have facilitated the use of digital elevation model (DEM) and DEM of difference (DoD) methods for studying geomorphic processes in complex terrain. However, insufficient understanding of systematic bias and random errors for DEMs constrained the application. In this study, we comprehensively analyzed the spatial pattern and magnitude of errors (including systematic and random errors) of DEMs derived from RAV-acquired point clouds for a topographically complex area (a subcatchment of Qiaogou in the hilly and gully loess plateau (SC_QG), China). The relationships between random errors and influential factors associated with topography, point cloud density, vegetation, and interpolation algorithms were also evaluated. On this basis, an error source thresholding (EST) method was adapted through incorporating residual systematic errors and including more impacting factors in the fuzzy inference system for random error estimation. The adapted EST (AEST) method was then employed to quantify the DoD uncertainty and geomorphic changes in two small catchments with complex terrain (i.e., SC_QG and a sub-catchment of Telagou (SC_TLG) in the hilly and gully Loess Plateau, China), while the results were verified by the changes measured by terrestrial laser scanning (TLS) and erosion pins, respectively. Results showed that mean value of systematic errors of DEMs were 0.065 and 0.005 m for SC_QG and SC_TLG, while the residual errors were reduced to 0.002 and 0.001 m after co-registration, respectively. Significant statistical relationships ($p < 0.01$) were found between random errors and influential factors. The erosional volume of two study sites detected by the adapted method were −252.29 and −981.07 m^3 and the corresponding depositional volume were 30.57 and 1594.32 m^3, respectively. The adapted method achieved a comparable pattern and magnitude of volumetric changes with TLS results, which was superior to the original EST method in SC_QG. Besides, our method showed a lower absolute error (0.034 m) compared to the original method (0.087 m) through a comparison with erosion pins measurement in the SC_TLG. Overall, the AEST method provided a reliable tool for geomorphic change detection in areas associated with complex terrain.**

***Index Terms*— 3-D point clouds, digital elevation model (DEM) errors, fuzzy inference system (FIS), steep slopes, topographic changes.**

I. Introduction

LAND surface with complex topography (e.g., mountainous areas) has been a hotspot area for disasters relevant to geomorphic processes such as soil erosion [1], landslides [2], and debris flows [3]. However, topographically complex areas are usually difficult to reach, constraining the observation of the processes [4]. Recent advances in remote aerial vehicle (RAV) remote sensing technologies, including structure from motion (SfM) photogrammetry [5] and laser scanning (LS) [6] have dramatically facilitated the acquisition of high-resolution topographic data [7] and thus the detection of the surface processes on topographically complex areas.

Remote aerial vehicle laser scanning (ULS) is a powerful instrument for the analysis of geomorphic change in complex environments for its convenience, high accuracy, and capability of partly penetrating vegetation [8]. However, ULS typically produces relative low point density on steep-sloping areas, limiting its use in the complex terrain environment [9]. RAV photogrammetry (RAV-P) with multiview-stereo (MVS) algorithms have often been employed to generate dense point clouds [5], [10], while it was a visible light-based technique that is incapable of exploring terrain information on vegetated areas. Recently, considerable practices were conducted on fusing the point cloud data from different sources [11], [12], including fusion method development [13], 3-D reconstruction [14], and object detection [15]. The fusion of ULS data and RAV-P data promoted a more comprehensive and accurate representation of the details of steep topography [16], largely benefiting the monitoring of geomorphic processes on areas with complex terrain [17].

In addition to data sources, several algorithms have also been developed for geomorphic change detection (GCD),

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such as 3-D based methods multiscale model to model cloud comparison (M3C2) algorithm [18] or slice contraction change detection (SCCD) algorithm [19] and 2.5-D based digital elevation model (DEM) differencing [20]. The DEM of difference (DoD) methods required DEM as inputs, which could be produced based on a wider range of data sources such as dense point clouds obtained by terrestrial laser scanning (TLS) or relatively sparse point clouds acquired by airborne or space-borne remote sensing techniques. This has largely enhanced the flexibility of the application of DoD methods [21].

The generation of DEMs involved point cloud filtering (the process of removing nonground points) and DEM interpolation [22]. Numerous point cloud filtering algorithms have been developed to accurately extract ground point cloud data, including multiscale curvature classification (MCC) [23], cloth simulation filtering (CSF) [24], morphological algorithms [25], and progressive triangulation [26]. Li et al. [27] previously compared the applicability of different filtering algorithms on the Loess Plateau, including MCC, elevation threshold with expand window (ETEW), adaptive TIN (ATIN), progressive morphology (PM) and slope-based filtering (SBF) in complex terrain areas, demonstrating the robustness of the MCC algorithm for the complex terrain environment. In addition, the CSF algorithm performed well with comparable classification accuracy and efficiency to other algorithms [24]. Zeybek and Şanlıoğlu [28] demonstrated that the CSF algorithms showed the highest accuracy classifications when compared to the MCC, Triangular irregular network (TIN), and surface-based filtering by the use of RAV-P derived point cloud data in all scenarios. Furthermore, different DEM interpolation methods have been developed to convert the point clouds into continuous surface, including inverse distance weighting (IDW), TIN, and natural neighborhood (NN) [29]. Among those methods, the TIN method has been widely used for terrain interpolation thanks to its suitability for the representation of complex terrains [30]. However, the filtering and interpolation procedures inevitably introduce errors into the derived DEMs, eventually affecting the accuracy and reliability of DEMs.

DEM errors usually contain systematic and random errors [31], [32], which were then propagated into DoDs and eventually impacted the accuracy of change detection results [33]. The core philosophy of the DoD method was to distinguish DoD uncertainty from true geomorphic changes [34]. In previous studies, DoD uncertainty was often assessed using the level of detection (LoD), which was derived based on the errors of individual DEMs using the error propagation theory [33], [35]. Geomorphic changes that exceeded the derived DoD uncertainty were then considered as true changes [36]. In these methods, DEM errors were usually assessed using the global statistical analyses such as root mean square errors (RMSEs) derived from limited ground control points (GCPs) [37]. However, the spatial heterogeneity of DEM errors, particularly for complex terrain areas, was hardly expressed by the limited number of GCPs, which led to an inappropriate estimation of DEM errors and then inaccurate results of the subsequent change detection [38].

Recognizing this issue, considerable efforts have been made to establish methods for deriving spatial DEM errors. Some representative studies were as follows. Wheaton et al. [33] proposed the error-source thresholding (EST) method through linking various inputs parameters (e.g., slope and point density) with DEM errors based on the fuzzy inference system (FIS). Milan et al. [30] developed the variance-overcome thresholding (VOT) method based on the regressions between topographic roughness and DEM errors. In addition, Monte Carlo methods have also been employed to simulate the spatial variations of DEM errors [39]. Among those methods, the EST method was the most commonly adopted one for monitoring surface processes in different environments mainly due to the flexibility of input parameters [21], [40], [41]. Nevertheless, the FIS rules in the original EST method were established based on empirical experience rather than the quantitative relationship between DEM errors and influential factors [42]. This may constrain the reasonability of the rules and even leads to the loss of some important information (e.g., nonlinear relationship between DEM errors and impacting factors) [37]. Furthermore, the DoD methods did not consider the systematic bias between DEMs for different points in time [43]. However, recent studies have demonstrated that systematic biases (e.g., those induced by internal sensor parameters of instruments) between repeated surveys were prevalent in the context of multitemporal changes [44], [45], which may exert considerable impacts on subsequent results [34].

Here, we aimed to improve the accuracy of GCD in complex terrains through accounting for both systematic and random errors in the EST method. To this aim 1) the magnitude of systematic bias and residual errors after co-registration were assessed between the repetitive DEMs derived from the point clouds fused from ULS and RAV-P data acquired from one study site with complex terrain; 2) the regression relationship was established between the random errors and influential factors (including slope, topography roughness, point density, vegetation cover, and DEM interpolation errors); 3) the EST method was adapted through incorporating residual systematic errors and including more impacting factors in the FIS for random error estimation; and 4) the proposed method was employed to monitor soil erosion and deposition of two sites (one of them was for the development of the method). The reliability and performance of the adapted EST (AEST) method were also verified based on TLS results and erosion pin measurements.

II. Materials and Methodology

A. Materials Description

The Qiaogou catchment, covering an area of 0.45 km^2, is located in the hilly and gully Loess plateau, China [46] [see Fig. 1(a)]. Mean annual temperature and precipitation were approximately 9.9 °C and 467.6 mm, respectively, between 1957 and 2012, respectively [47]. The precipitation was mainly concentrated in the rainy senson (from July to September), in the form of localized short-duration rainstorms, often resulting in the severe soil erosion. The catchment was thus

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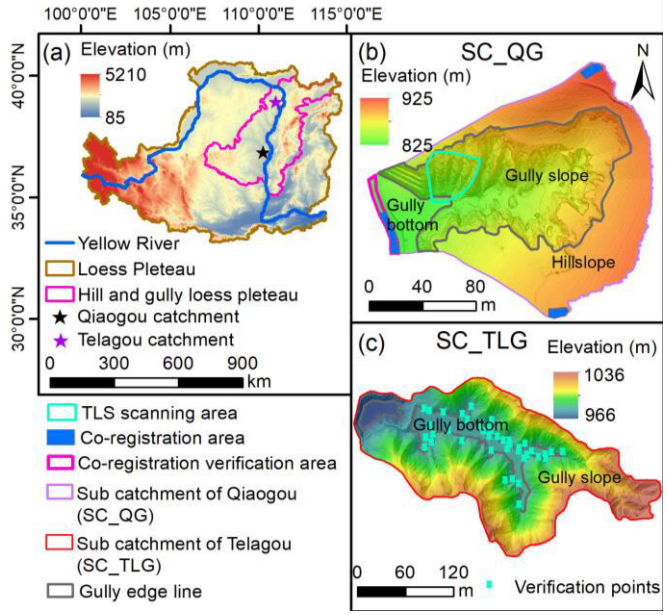


Fig. 1. (a) Overview of the study site, showing the location of Qiaogou catchment and Telagou catchment in the Loess Plateau. (b) DEM of a sub-catchment of Qiaogou (SC_QG) with TLS scanning area marked and geomorphic units, including hillslope, gully slope and gully bottom, differentiated by the gully edge line. (c) DEM of a SC_TLG.

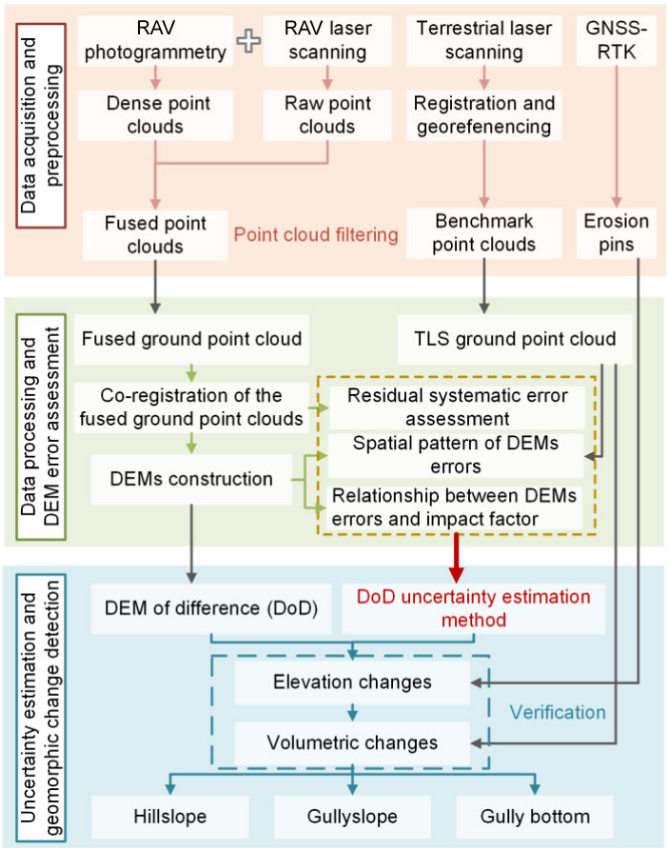


Fig. 2. Workflow of this study.

characterized by a fragmented and complex topography, comprising of hillslopes, gully slopes, and gully bottom [48]. The slope gradient of hillslopes is usually lower than 30°, while the coverage of vegetation, dominated by grass, is between 60% and 70%. Gully slopes are characterized by slope gradients in excess of 40°, with vegetation coverage that was commonly under 10%. Gully bottoms are topographically gentle with a few trees and shrubs distributed on both sides. Since 2001, the catchment has been naturally restored to control severe soil erosion, following the policy on returning slope croplands to grasslands and woodlands [49]. Various plant species have been established within the catchment, with dominant species including *Ziziphus sativa*, *Atemisia gmelini*, *Stipa bungeana*, *Heteropappus altaicus*, *and Lespedeza davurica* [47]. Soil erosion has been well controled by the restored vegetation, while erosion could still be induced by intensive rainfall events and led to geomorphic changes. In the study, a subcatchment of Qiaogou (SC_QG) was chosen for the development and validation of the AEST method [see Fig. 1(b)].

The Telagou catchment, covering an area of 128.2 km^2, is suitated in the northern part of the hilly and gully Loess Plateau, China [see Fig. 1(a)], which is a subbasin of the Huangfuchuan Basin and is considered as the most intensively eroded area of Loess Plateau [50]. The average annual temperature and precipitation are approximately 7.5 and 400 mm, respectively. Like Qiaogou, precipitation in Telagou also mainly occurred as rainstorms in the rainy senson (from July to September) [51]. The geomorphic units consist of hillslopes, gully slopes, and gully bottom. The soil types in the basin are *pisha* sandstone, loess, and wind sandy soil. Due to the sparse vegetation cover, the *pisha* sandstone was largely exposed in the study site. The large terrain elevation difference and storm generated run-off scouring lead to serious soil erosion. Therefore, numberous engineering measures (e.g., cherk-dams) have been constructed in the basin to mitigate the soil erosion [51]. In this study, a subcatchment of Telagou (SC_TLG) was chosen to validate the tranferability of the AEST method [see Fig. 1(c)].

B. Methodology

This study included three parts as shown in Fig. 2.

1) Point cloud data and photogrammetry data of two study areas were acquired by the ULS and RAV-P instruments. The dense point clouds derived from RAV-P images was merged with ULS-derived point clouds to produce the fused point clouds. TLS results and erosion pins measurements were employed as the validation data and were acquired simultaneously with RAV datasets. All the point cloud datasets underwent automatic filtering and manual filtering processes to derive the bare ground point clouds.
2) The co-registration process was undertaken between the fused point clouds to reduce the systematic bias, while residual systematic bias were then assessed for the co-registered point clouds. Subsequently, the quality of DEMs derived based on the co-registered point clouds were evaluated based on TLS ground points. The regressions between DEM errors and influencing factors, including slope, point density, vegetation cover,

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TABLE I
BASIC PARAMETERS OF ULS SURVEYS FOR TWO STUDY SITES

Study site	SC-QG	SC-TLG
Parameter	ULS	ULS
Equipment	SZT-R250	SZT-R250
Flight dates	July 2021, November 2021	July 2022, March 2023
Flight height (m)	70	100
Flight speed (km/h)	7	7
Scanning angle (°)	90°- 270°	90°- 270°
Pulse (kHZ)	100	100
Scanning (line/s)	100	100

and roughness, were also established based on datasets acquired from the SC_QG.

3) On this basis, an adapted DoD uncertainty estimation method was developed to derive the DoD uncertainty and quantify the spatial pattern of erosional and depositional changes of different geomorphic units (hillslope, gully slope, and gully bottom) in two study sites. The geomorphic changes derived from the adapted method were verified by the TLS measurements in SC_QG and erosion pins measurement (elevation changes) in SC_TLG, respectively.

1) Data Acquisition:

a) Remote aerial vehicle laser scanning: The ULS surveys were performed in July 2021 and November 2021 for SC_QG and in July 2022 and March 2023 for SC_TLG (see Table I). An intersecting-parallel flight route was taken to fully capture topographic information of the study sites. The detailed RAV flight settings and LS parameters were summarized in Table I. All the point cloud datasets were georeferenced and projected in the same reference system of China Geodetic Coordinate System 2000 (CGCS2000). Due to the range of the ULS equipment, it failed to capture the full hillslope of SC_TLG.

b) RAV-photogrammetry: A Phantom 4 Pro v2.0 RAV platform carrying an FC6310S digital camera (8.8 mm focal length and 12.4 million effective pixels) was employed to acquire images of SC_QG and SC_TLG, at dates that were consistent with those of ULS data collection. The flight height and flight speed of the RAV-P equipment were 100 m (120 m for Telagou) above the tableland and 10 km/h, respectively. The RAV data were acquired from five different angles (detailed see [9]) with the forward and side overlap of 80% and 70%, respectively. Additionally, four GCPs in the SC_QG and SC_TLG were placed and measured using global navigation satellite system real-time kinematic (GNSS-RTK) following the CGCS2000 system, with an average horizontal and vertical accuracy of 0.01 and 0.02 m, respectively. Subsequently, the ContextCapture Center 4.4.10 software was employed to produce the dense point clouds based on the photographs taken, with the processing details presented in Yan et al. [9]. Due to the same reason as ULS, the RAV-P was also failed to collect the full hillslope of SC_TLG.

c) Terrestrial laser scanning: A Leica ScanStation C10 scanner was employed to scan a slope of SC_QG [see Fig. 1(b)] in July 2021 and October 2021, on the same dates as the RAV flights. The scanner offers the highest scanning rate of 50 000 points per second, with a single and target scanning accurate of 6 and 2 mm, respectively [44]. The TLS was set up in four different locations for fully capturing the topographic information of the slope. Three sphere targets were placed in the fixed position around the slope to ensure that they were always within the range of the TLS scanning at different locations. The 3-D coordinates of the fixed position were measured in the CGCS2000 coordinate system. The point clouds of different scans were co-registered to form a complete data of the slope based on the sphere targets. Finally, TLS point clouds and their resulting geomorphic change were derived as the benchmark for assessing DEM errors and quantifying the accuracy of change detection results derived from RAV datasets.

d) Erosion pins: Due to the complex topography of SC_TLG, TLS was incapable of fully capturing the complete region. Therefore, erosion pins were selected as the validation data. Sixty erosion pins were placed throughout this study site, with each pin being 1.2 m, of which the depth of burial was 0.6 m. As the topography of gully slope was steep, most of the erosion pins were distributed uniformly at gully bottom, with a small number of pins located in the accessible gully slope area. The erosion pins were buried in July 2022, coinciding with the date of RAV flights. GNSS-RTK was employed to measure the location of the erosion pins in the CGCS2000 coordinate system. During the second RAV flight period (March 2023), the heights of exposed portions of erosion pins were measured using tape rules. However, some of the erosion pins were destroyed during this period due to the scouring from storm-generated runoff, and forty erosion pins elevation changes were ultimately retained as validation data [see Fig. 1(c)].

2) Data Processing:

a) Point cloud filtering: The (MCC) filtering algorithm, proposed by Evans and Hudak [23], was employed to classify and extract the bare ground points from the ULS and TLS dataset. This algorithm has been demonstrated to be suitable for complex topography [52] and has also been used in the study area [48] Additionally, the CSF algorithm, developed by Zhang et al. [24], was used for the classification of the dense point cloud of RAV-P, with the parameters being set with reference to Yan et al. [9]. Considering the complexity of the terrain in the study area, the automatic classification was not entirely accurate, especially in steep areas. Therefore, the results of filtering algorithms were also manually checked and adjusted in the Terrasolid software to further improve the filtering accuracy (http://terrasolid.fi).

b) Point cloud co-registration: Although the ULS point cloud datasets were georeferenced using the uniform coordinate system, the misalignment still existed, caused by factors such as the slope, aspect, and roughness. The point clouds for different dates were thus co-registered by the iterative closest

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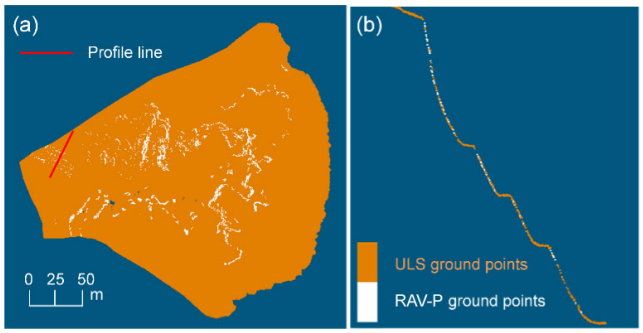


Fig. 3. (a) Top view of fused ground point clouds and (b) aside view of the profile taken from the fused point clouds for SC_QG.

point (ICP) algorithm [53] based on the stable areas (portion of the hardened road and bottom of the transmission tower) in the study area (see Fig. 1). The validation area (see Fig. 1) was also chosen to assess the reliability of co-registration results. Subsequently, the ICP algorithm was also employed to co-register the RAV-P point clouds, with the registered ULS point cloud dataset as a reference. This reduced the systematic bias between RAV-P point clouds as well as those between ULS point clouds and RAV-P point clouds.

c) Fusion of ULS and RAV-P derived bare ground points: Despite taking the intersecting parallel flight routes, it was difficult to fully capture the complex terrain information, especially for the steep slopes. Therefore, the RAV-P-derived dense point cloud was employed to fill the blank areas of the ULS point cloud. Considering the high slope and lack of vegetation cover in steep areas, the gap areas were defined as regions where slope gradient was over 70° and fractional vegetation cover (FVC) was less than 20%. The derivation of slope gradient and FVC is described in Section II-B2e. Subsequently, the dense ground point derived by RAV-P were subsampled with reference to the average point density of the ULS-derived ground points. This process was done to facilitate the fusion of ground point clouds and following statistical analysis. Finally, the fused point clouds was obtained (see Fig. 3).

d) DEM construction: Point cloud data were interpolated to generate DEMs for subsequent terrain change and hydrological analyses [54], [55], while point cloud density is a crucial factor that determines the resolution of generated DEMs [56]. Given the mean point density of ground points acquired in the study (118–157 points $\cdot$ m^{-2}), the 0.1 m resolution DEMs were produced using the TIN interpolator based on the fused point clouds of the SC_QG and SC_TLG, respectively. Previous studies have indicated that this method returned lower elevation errors than other interpolation methods [57].

e) DEM error assessment: The DEM errors ($e(x)$) normally consisted of systematic errors (μ) (or called bias) and random errors [$\varepsilon(x)$]

$$e(x) = \mu + \varepsilon(x). \tag{1}$$

Previous studies commonly employed global statistical analysis to evaluate the precision (random errors) in DEMs, including RMSE or standard deviation (STD) of errors at the location of the ground check points [37], [58]. In this study, RMSE and STD were also initially utilized to assess the precision in DEMs of the fused point clouds, taking the TLS-derived ground points as reference data. The RMSE and STD were calculated as

$$\text{RMSE} = \sqrt{\sum \left(z_{\text{DEM}_i} - z_{\text{reference}_i}\right)^2 / n} \tag{2}$$

where z_{DEM_i} was the elevation value of DEMs interpolated from the fused ground point cloud. $z_{\text{reference}_i}$ was the z-coordinate of TLS point clouds, n is the number of the TLS points

$$\text{STD} = \sqrt{\frac{\sum_{i=1}^{n} \left(Z_{\text{DEM}_i} - \overline{Z_{\text{DEM}}}\right)^2}{n-1}} \tag{3}$$

where $\overline{z_{\text{DEM}}}$ was mean elevation of the DEMs derived from fused point clouds.

We also quantified the cell-by-cell relationship between random error and DEM slope, topography roughness, point density, and vegetation related parameter, respectively. The DEM slope in degrees was expressed in terms of the maximum rate of change within a 3 × 3 cell neighborhood [42]. Topographic roughness was employed via the roughness index-elevation, proposed by Cavalli et al. [59], which was used for LiDAR-derived point cloud [60]. Point densities were calculated using a localized spherical neighborhood of 1 m radius implemented in CloudCompare v2.10 software instead of surface densities, ensuring logical point densities in steep areas. FVC, derived from the visible light difference vegetation index (VDVI) using the RAV-P dataset, was employed to express the vegetation coverage [61].

In order to further analyze these relationships, the slope gradient was divided into nine levels with the 10° as an interval, and the point density was divided into 14 levels with the 10 points/m^2 as an interval. The FVC was similarly categorized into ten levels with 10% of as an interval. Finally, the topographic roughness was classified as 11 grades at a 0.1 m interval and an extra level of >1 m. The STD of the DEM errors was then calculated for the range corresponding to each level of the slope, point density, FVC, and topographic roughness, respectively. The quantitative relationships between STD of DEM errors and influential factors were thus established.

The systematic error (μ) in the DEMs from the fused ground points was also determined before performing the change detection analysis. The systematic error was derived as the average distance ($\hat{\mu}$) between the fused point clouds of the paved roads (without topographic change) of July 2021 and November 2021 in SC_QG and buildings of July 2022 and March 2023 in SC_TLG, including the distance before and after the co-registration. The distance between the two point clouds was calculated using the M3C2 algorithm [18] implemented in CloudCompare v2.10 software

$$\hat{\mu} = \overline{Z_1} - \overline{Z_2}. \tag{4}$$

$\overline{Z_1}$ and $\overline{Z_2}$ were the mean elevation value of fused ground points.

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3) Geomorphic Change Detection:

a) DEM of difference: The principle of the raw DoD follows the following equation:

$$\mathrm{DoD} = \mathrm{DEM}_1 - \mathrm{DEM}_2 \tag{5}$$

where DEM_1 and DEM_2 are the DEMs derived from the fused point clouds acquired at different points in time, respectively,

$$\mathrm{DoD}_{\mathrm{actual}} = \mathrm{DoD} \pm \delta_U \tag{6}$$

where $\mathrm{DoD}_{\mathrm{actual}}$ is the true topographic changes and δ_U is the DoD uncertainty.

The commonly adopted framework for quantifying DoD uncertainty was proposed by Brasington et al. [35], which combined each of DEM errors using the error propagation theory

$$\delta_U = t\sqrt{(\delta_{Z_1})^2 + (\delta_{Z_2})^2} \tag{7}$$

where δ_{Z_1} and δ_{Z_2} are the STD (or RMSE) of DEM errors (random errors, assuming a Gaussian distribution) in fusion point cloud data for each DEM, respectively. The value of t equals to 1.96, normally representing a 95% confidence interval [20], [62].

b) Adapted method for DoD uncertainty estimation: For the original EST method, the input parameters were only DEM-derived slope and point density in the FIS [33]. The spatial pattern of DEM errors was then implemented in the GCD software (http://gcd.joewheaton.org). Based on the assumption of a Gaussian distribution random errors. Equation (7) was employed to obtain the corresponding DoD uncertainty.

Given that the errors were composed of systematic errors and random errors, the AEST was then proposed

$$\delta_{U_{\mathrm{DoD}}} = 1.96 * \sqrt{(\delta_{Z_1})^2 + (\delta_{Z_2})^2} + \sigma \tag{8}$$

where the σ represents the residual systematic error after co-registration of fusion point cloud data, which was derived from the mean distance (using vertical normal scale of M3C2 method) of different periods point cloud in the validation area.

In the study, slope gradient, roughness, point density, and FVC were considered as influential factors on DEM errors. Besides, the DEM construction process imposed the interpolation error, which was propagated into DoD uncertainty and was thus therefore additionally taken into account in our study. The interpolation errors derived from DEM interpolation method was defined as the differences between the DEM elevation and the Z-value of the fused point cloud datasets.

The new FIS rules (see Table II) and the membership functions were set in terms of the empirical relationships between STD of random errors and slope, roughness, FVC and point density. However, we did not derive a regression relationship between the interpolation errors and the STD of random errors. The interpolation error using TIN algorithm is normally low [57], especially using the quite high-density (over 100 points/m^2) point cloud datasets. While high interpolation error value usually exhibited in the local steep wall due to the omission of the points. The interpolation errors membership function was thus defined two membership functions (low and extreme) with a low level showing the majorly low interpolation error and the extreme levels representing the local and sparse high interpolation errors [42].

4) Quantification of Erosion and Deposition Volume: The volume of erosion and deposition was calculated as follows:

$$\mathrm{V} = \sum_{1}^{n} d_i \times S \tag{9}$$

where V is volumetric change of erosion or deposition (m^3), n is the total number of grids, and d_i is the difference in elevation of the grid (m) and S is the area of the grid (m^2). Subsequently, the volumetric net change (V_0) was thus derived from the following equation:

$$V_0 = V_e - V_d \tag{10}$$

where V_e is the erosion volume and V_d is the deposition volume, respectively.

5) Verification of the Volumetric Changes: To determine the reliability of the volumetric change in the SC_QG, it was compared against TLS-derived volumetric changes on the chosen slope [see Fig. 1(b)]. The M3C2 algorithm [18] was adopted to derive the erosion and deposition volume for the selected slope based on dense TLS point clouds. The M3C2 algorithm estimated the topographic change directly based on the 3-D point clouds, through calculating a normal vector (D) for each point and defining a cylinder of radius (d) following the direction of the normal vector. The average distance between two TLS point clouds in the defined cylinder (M3C2 distance) was considered as the topographic changes. The multiscale normal was chosen to derive the M3C2 distance, with D and d being 0.2 and 0.1 m, respectively.

In addition, topographic changes in the SC_TLG were verified using erosion pins measurements [see Fig. 1(c)]. The mean absolute errors (MAEs) were applied to assess the adapted method reliability and robustness

$$\mathrm{MAE} = \frac{\sum(C_U - C_e)}{n} \tag{11}$$

where C_e is the elevation change of each of the erosion pins; C_U is elevation change for each cells corresponding to the location of erosion pins, and n is the number of erosion pins.

III. Results

A. Systematic Bias Assessment

Spatial pattern of systematic bias was considerable different for two study sites (see Fig. 4). The SC_QG was found to have a significant systematic bias, with the mean value and median value between the original fused point clouds of the cement-paved road of being −0.065 and −0.063 m, respectively. In contrast, the mean value and median value dramatically decreased to −0.002 to 0.001 m after the co-registration (see Fig. 5). However, the stable validation area (the top of the building) in the SC_TLG showed slight systematic bias between the different periods of point cloud dataset, with mean value and median value being −0.005 and −0.004 m, respectively. After the co-registration process, the mean and median values of point cloud dataset distance were both decreased to 0.001 m.

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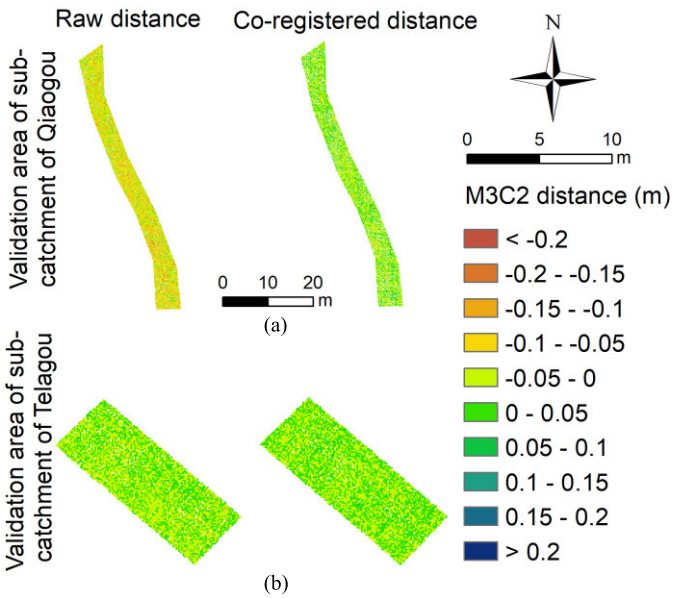


Fig. 4. Spatial pattern of systematic bias and residual bias after co-registered for (a) SC_QG and (b) SC_TLG.

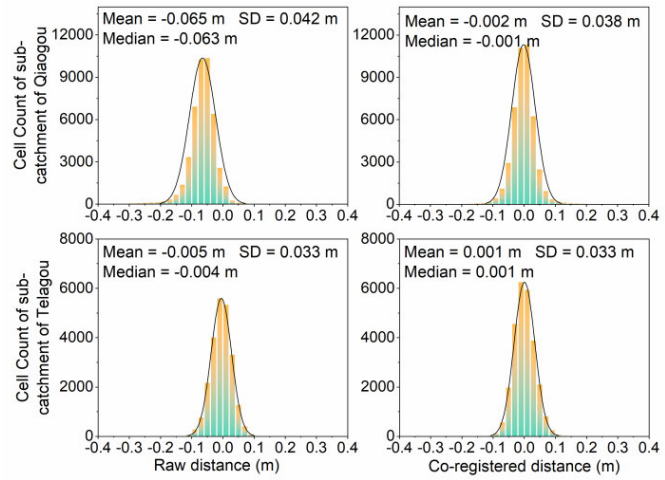


Fig. 5. Histogram of systematic bias and residual bias corresponding to Fig. 4.

B. Random Error Analysis

The random errors showed similar spatial patterns and approximate Gaussian distribution in the selected slope [see Fig. 1(c)] between the period of July 2021 and November 2021 for the SC_QG (see Fig. 6). Meanwhile, the random errors (magnitude typically exceeding 1 m) in steep topography were far beyond those in other areas with relatively gentle topography (Fig. 6 marked by black rectangular). Among them, the RMSE of the random error were 0.225 and 0.216 m in July 2021 and November 2021, respectively. The STD were on the same order of magnitude with RMSE, which were 0.225 and 0.211 m, respectively.

The scatter plots of slope gradients and DEM random errors showed that the magnitude and fluctuation of random errors were lower for the slope gradient lower than 25° [see Fig. 7(a)]. While the slope degree was over 70°, random errors dramatically dispersed. With the increase in point density, the

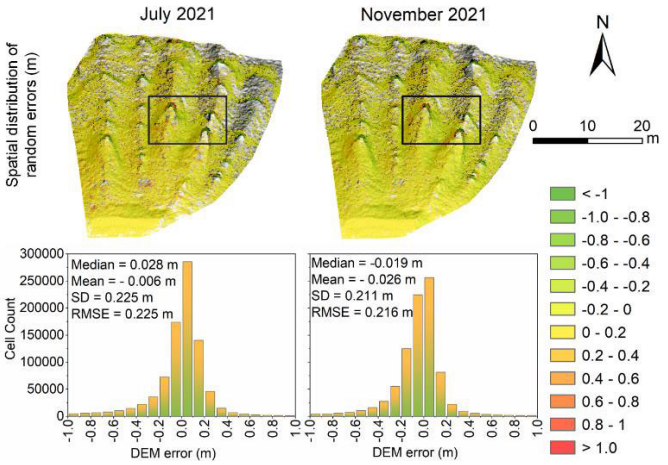


Fig. 6. Spatial pattern and cell count with different grades of random error for the selected slope of SC_QG in July 2021 and November 2021.

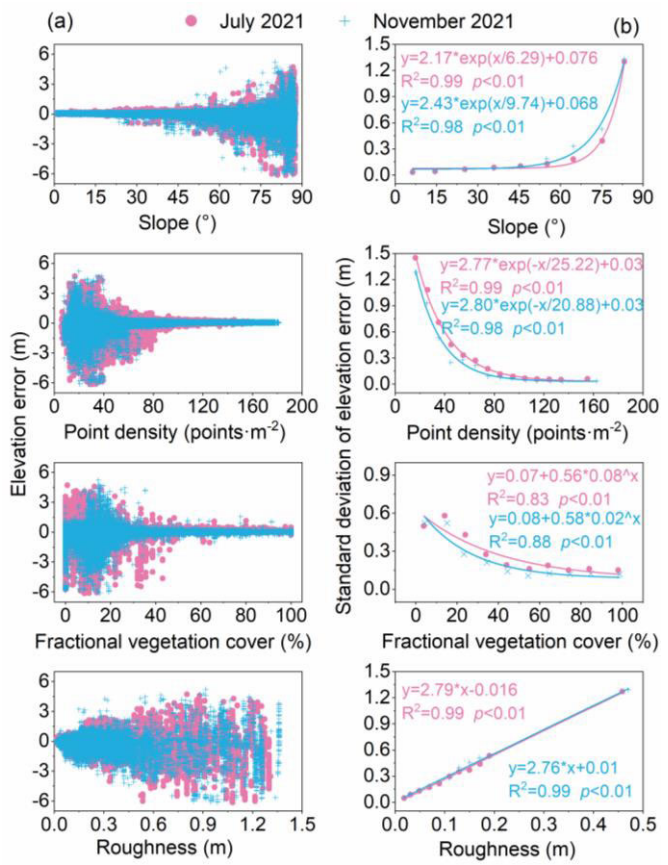


Fig. 7. Scatterplot and empirical relationship between different influencing factors and random errors for the selected slope (a) in July 2021 and (b) November 2021 of the SC_QG.

random error decreased sharply and the fluctuation gradually stabilized. The relationship between the random error and FVC indicated similar trend with random errors and point density. Additionally, the error also enlarged gradually with increasing of roughness.

A significantly positive exponential correlation was found between slope and STD of DEM errors, with R^2 of 0.99 ($p < 0.01$) and 0.98 ($p < 0.01$) in July 2021 and November 2021, respectively [see Fig. 7(b)]. The topographic roughness

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and the STD of DEM errors showed a significant linear relationship ($R^2 = 0.99$ $p < 0.01$). However, the point density and STD of DEM errors demonstrated a significant negative exponential relationship, with R^2 of 0.99 ($p < 0.01$) and 0.98 ($p < 0.01$) in July 2021 and November 2021, respectively. The FVC showed a similar trend with point density ($R^2 = 0.83$ $p < 0.01$ and $R^2 = 0.88$ $p < 0.01$). Finally, these empirical relationships were also referenced for the FIS model construction.

C. Quantification of DoD Uncertainty and Estimation of Erosion/Deposition

Following the results in Fig. 7(b), the membership function of point density was divided from three levels including low level (0–40 points/m^2), middle level (35–110 points/m^2), and high level (100–1000 points/m^2). Three levels of DEM uncertainty (STD of random error) were thus achieved, including 0–0.05, 0.05–0.45 m and greater than 0.45 m. The levels of DEM uncertainty were then used to derive the grades of slope and roughness. Slope gradients was also classified into three levels, including low level (0°–25°), middle level (20°–75°) and high level (70°–90°) (see Fig. 8). The roughness was divided into 0–0.025, 0.02–0.2 m and over 0.19 m. Due to the roughness was mostly within 0.02–0.2 m, we additionally separated this rank into two parts (0.02–0.1 and 0.09–0.20 m) in more detail the effect of roughness on the derived DEM uncertainty. Topographic roughness were thus formed four levels, including low level (0–0.025 m), middle level (0.02–0.1 m), high level (0.09–0.2 m), and extreme level (>0.19 m). Otherwise, the FVC was categorized into three ranks, which were low rank (0–0.2), middle rank (0.1–0.6), and high rank (0.6–1) following the relationship between FVC and STD of DEM errors. The interpolation errors were divided into two levels, including a low level (0–0.3 m) and an extreme levels (>0.2 m). On the basis of roughness levels, the DEM uncertainty was also classified into four levels, including low level (0–0.05 m), middle level (0.04–0.20 m), and high level (0.18–0.45 m) and an extreme level (0.4–5 m).

On the basis of the fuzzy membership functions, the DEM uncertainty were thus derived and been employed to derive the DoD uncertainties (see Fig. 9). The raw DoD showed that topographic changes (erosion and deposition) occurred throughout the two study sites and the percentage of area for raw DoD showed an approximately normal distribution, with highest percentage being −0.1 to 0.1 m for two study sites (see Fig. 9). DEM error-derived DoD uncertainty were notably larger on the gully slope than that on the hillslope and gully bottom. For the SC_QG, the majority area of gully slope were subject to a DoD uncertainty between 0.3 and 0.4 m, with localized extreme value (over 0.5 m) occurring in the steep terrain, while the gully bottom and hillslope exhibited the lowest magnitude of uncertainty (0–0.1 m). For the SC_TLG, more than 80% of the DoD uncertainty was in the range of 0–0.1 m. The true elevation change was remarkably different from the raw DoD, with erosion and deposition process occurring in portions of the study area. Soil erosion mainly occurred on gully slopes and deposition normally appeared in gully bottom. However, the two study sites showed considerable difference in area distribution of

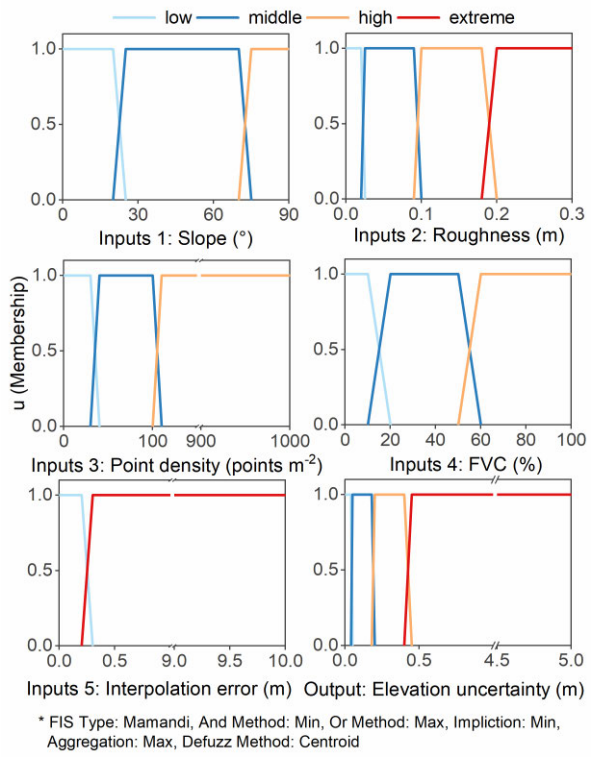


Fig. 8. Fuzzy membership functions of input and output parameters and FIS fundamental information adopted in the study.

geomorphic changes (see Fig. 10). The highest percentage of erosion area was in the range of −0.2 to −0.1 m, and highest percentage of deposition area was in the range of 0–0.1 m for the SC_QG. While the distribution was considerably different in the SC_QG, with highest percentage of erosion area being in the level of lower than −0.5 m and highest percentage of deposition area being in the range of 0.1–0.2 m.

The volume of erosion (−252.29 m^3) exceeded the volume of deposition (30.57 m^3) throughout the SC_QG, while the results of SC_QG were on the contrary (see Table III). For the SC_QG, gully slope presented the highest percentage of erosion (72.4%) and deposition (48.6%), followed by the hillslope (24.3% of erosion and 34.4% of deposition). The lowest magnitude of erosion (3.3%) and deposition (17%) could be detected on gully bottom. The magnitude of volumetric changes in the SC_TLG extended much higher than the SC_QG, with total erosion volume of −981.07 m^3 and deposition volume of 1594.32 m^3, respectively. The vast majority of erosion (99.7%) occurred on gully slopes due to the lack of hillslope information, while deposition occurred mainly at gully bottom (66.9%) and sporadically in gully slopes (33.1%).

D. Validation of RAV Point Clouds Derived-Change

Erosional changes were found to occur predominantly on gully slopes, whereas deposition was primarily located on the relatively flat area at the bottom of the gully slope (see Fig. 11). Notably, the raw method is flawed, with slight depositional change being ignored at both test sites [the area marked by the rectangular in Fig. 11(a) and (b)], while the

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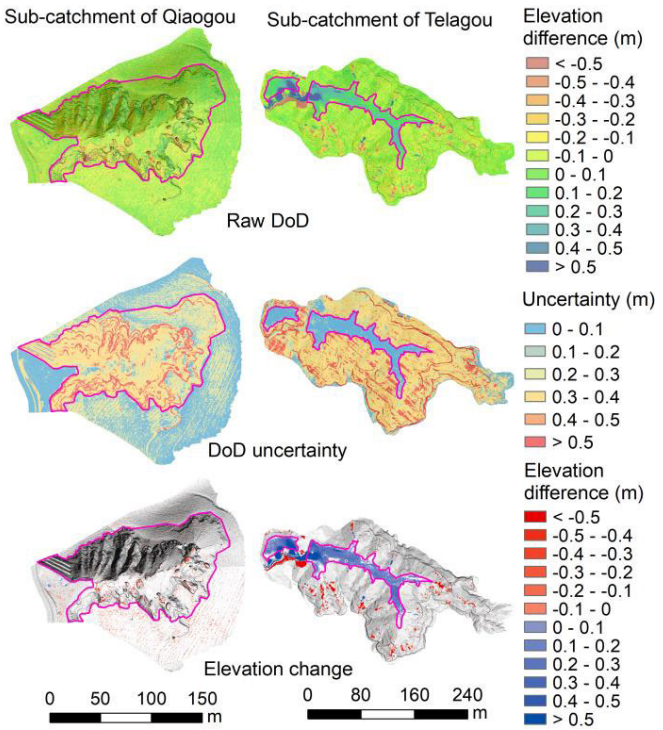


Fig. 9. Spatial pattern of raw DoD, DoD uncertainty, and elevation change.

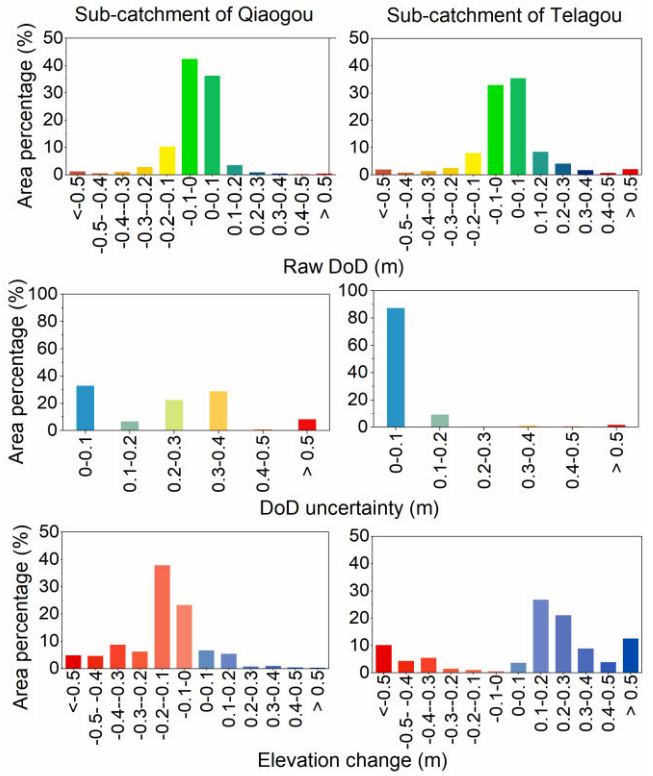


Fig. 10. Area percentage of ranges for raw DoD, DoD uncertainty, and elevation change corresponding to.

adapted method could capture those details of slight depositional change [see Fig. 11(c) and (d)]. Spatial pattern of elevation changes using the AEST method generally resembled

TABLE II
INPUT–OUTPUT RULES USED IN FIS

Rule	Input					Output
	Slope (°)	Roughness (m)	Point density (pts/m^2)	FVC (%)	Interpolation Error (m)	δ_U (m)
1	Low	Low	Low	Low	NA	Low
2	Low	Low	Low	Medium	NA	Low
3	Low	Low	Low	High	NA	Low
4	Low	Medium	Low	Low	NA	Medium
5	Low	Medium	Low	Medium	NA	Medium
6	Low	Medium	Low	High	NA	Medium
7	Low	High	Low	Low	NA	Extreme
8	Low	High	Low	Medium	NA	High
9	Low	High	Low	High	NA	High
10	Low	Low	Medium	Low	NA	Low
11	Low	Low	Medium	Medium	NA	Low
12	Low	Low	Medium	High	NA	Low
13	Low	Medium	Medium	Low	NA	Low
14	Low	Medium	Medium	Medium	NA	Low
15	Low	Medium	Medium	High	NA	Low
16	Low	High	Medium	Low	NA	High
17	Low	High	Medium	Medium	NA	Medium
18	Low	High	Medium	High	NA	Medium
19	Low	Low	High	Low	NA	Low
20	Low	Low	High	Medium	NA	Low
21	Low	Low	High	High	NA	Low
22	Low	Medium	High	Low	NA	Low
23	Low	Medium	High	Medium	NA	Low
24	Low	Medium	High	High	NA	Low
25	Low	High	High	Low	NA	High
26	Low	High	High	Medium	NA	Medium
27	Low	High	High	High	NA	Medium
28	Medium	Low	Low	Low	NA	Medium
29	Medium	Low	Low	Medium	NA	Medium
30	Medium	Low	Low	High	NA	Medium
31	Medium	Medium	Low	Low	NA	High
32	Medium	Medium	Low	Medium	NA	Medium
33	Medium	Medium	Low	High	NA	Medium
34	Medium	High	Low	Low	NA	Extreme
35	Medium	High	Low	Medium	NA	High
36	Medium	High	Low	High	NA	High
37	Medium	Low	Medium	Low	NA	Low
38	Medium	Low	Medium	Medium	NA	Low
39	Medium	Low	Medium	High	NA	Low
40	Medium	Medium	Medium	Low	NA	Medium
41	Medium	Medium	Medium	Medium	NA	Medium
42	Medium	Medium	Medium	High	NA	Medium
43	Medium	High	Medium	Low	NA	High
44	Medium	High	Medium	Medium	NA	Medium
45	Medium	High	Medium	High	NA	Medium
46	Medium	Low	High	Low	NA	Low
47	Medium	Low	High	Medium	NA	Low
48	Medium	Low	High	High	NA	Low
49	Medium	Medium	High	Low	NA	Medium
50	Medium	Medium	High	Medium	NA	Low
51	Medium	Medium	High	High	NA	Low
52	Medium	High	High	Low	NA	High
53	Medium	High	High	Medium	NA	Medium
54	Medium	High	High	High	NA	Medium
55	High	Low	Low	Low	NA	High
56	High	Low	Low	Medium	NA	High
57	High	Low	Low	High	NA	High
58	High	Medium	Low	Low	NA	High
59	High	Medium	Low	Medium	NA	High
60	High	Medium	Low	High	NA	High
61	High	High	Low	Low	NA	Extreme
62	High	High	Low	Medium	NA	Extreme
63	High	High	Low	High	NA	Extreme
64	High	Low	Medium	Low	NA	High
65	High	Low	Medium	Medium	NA	Medium
66	High	Low	Medium	High	NA	Medium
67	High	Medium	Medium	Low	NA	High

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TABLE II
(Continued.) INPUT–OUTPUT RULES USED IN FIS

68	High	Medium	Medium	Medium	NA	Medium
69	High	Medium	Medium	High	NA	Medium
70	High	High	Medium	Low	NA	Extreme
71	High	High	Medium	Medium	NA	High
72	High	High	Medium	High	NA	High
73	High	Low	High	Low	NA	High
74	High	Low	High	Medium	NA	Medium
75	High	Low	High	High	NA	Medium
76	High	Medium	High	Low	NA	High
77	High	Medium	High	Medium	NA	Medium
78	High	Medium	High	High	NA	Medium
79	High	High	High	Low	NA	Extreme
80	High	High	High	Medium	NA	High
81	High	High	High	High	NA	High
82	NA	NA	NA	NA	Medium	Extreme
83	NA	Extreme	NA	NA	NA	Extreme

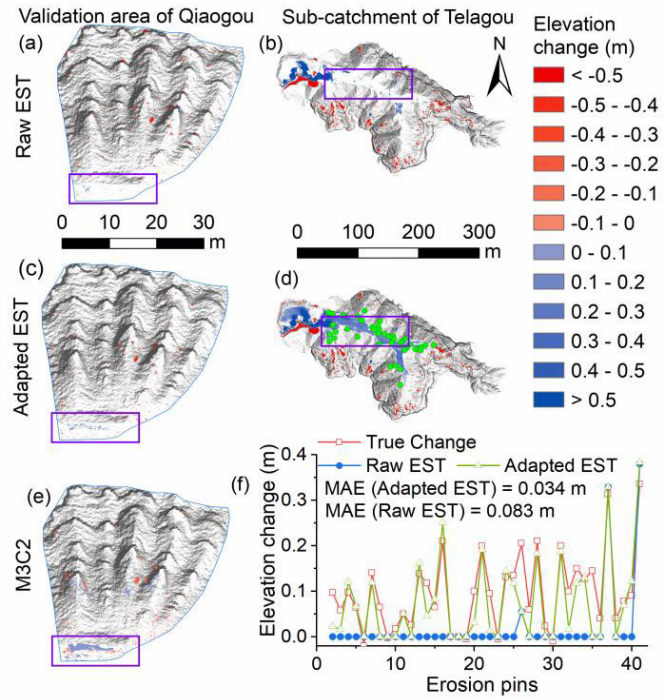


Fig. 11. Spatial pattern of elevation changes derived from (a) raw EST method, (c) AEST method and (e) M3C2 method on the selected slope for the SC_QG and spatial pattern of elevation changes derived from (b) raw EST method, (d) adapted method and (f) validation results of erosion pins for the SC_TLG.

that of the M3C2-derived result in the validation area of SC_QG. However, the adapted method captured less extent of erosion and deposition compared to the results of M3C2 method based on TLS point cloud in the validation area of SC_QG.

The volumetric changes (see Table IV) derived by the original EST method (erosion of -6.28 m^3 and deposition of 1.86 m^3), the AEST method (erosion of -3.29 m^3 and deposition of 0.65 m^3), and M3C2 method (erosion of -1.91 m^3 and deposition of 1.01 m^3) were on the same order of magnitude in the validation area of SC_QG, which were much smaller than the raw DoD (erosion of 86.65 m^3 and deposition of 28.33 m^3). The AEST method showed more accurate results using M3C2 results as the baseline, while overestimating the erosion volume and underestimating deposition volume for the selected gully slope. Besides, the MAE of AEST method and original EST method was 0.034 and 0.083 m, respectively [see Fig. 11(f)], based on the elevation changes of erosion pins [see Fig. 11(d)] in the SC_TLG. The adapted method performed better than the original EST method at both study sites.

TABLE III
VOLUMETRIC CHANGES AND PERCENTAGE OF DIFFERENT GEOMORPHIC UNITS, INCLUDING HILLSLOPE, GULLY SLOPE, AND GULLY BOTTOM

Study site	SC_QG		SC_TLG	
Geomorphic units	Erosion volume (m^3) and ratio (%)	Deposition volume (m^3) and ratio (%)	Erosion volume (m^3) and ratio (%)	Deposition volume (m^3) and ratio (%)
Hillslope	-61.19 (24.3%)	10.52 (34.4%)	-	-
Gully slope	-182.63 (72.4%)	14.86 (48.6%)	-978.11 (99.7%)	527.7 (33.1%)
Gully bottom	-8.41 (3.3%)	5.2 (17%)	-2.96 (0.3%)	1066.62 (66.9%)
Total	-252.29	30.57	-981.07	1594.32

IV. DISCUSSIONS

A. Spatial Variation of DEM Error and Systematic Bias

Assessment of DEM accuracy (systematic bias) and precision (random errors) enabled understanding the main factors affecting the DEM errors [58] and thus improved the equipment and data acquisition schemes, especially for a site where point cloud dataset derived DEMs were key to monitoring elevation changes [37]. Our results showed a noticeable systematic bias (0.065 m) in the point cloud data from different epochs (see Fig. 4). Although the systematic bias was less than that reported in relation to airborne LiDAR systematic bias (0.4–5 m) [41], it also affected the results of topographic change detection [34], especially when the changing signals were on a proximity order of magnitude to the systematic bias. Identifying and mitigating systematic biases would therefore meaningfully improve change detection results. At a simple level, systematic bias can be removed by moving one DEM to align it with another [41]. However, in steep terrain environments, slight horizontal shifts could cause severe vertical deviations [63]. ICP co-registration is able to spatially transform the dataset and has been widely adopted in the data registration process [34]. Notably, ICP registration required taking the presence of stable topographic areas (hardened roads, buildings, reefs) in the study area into account to ensure reliable alignment results as well as to assess possible residual systematic errors after the registration.

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TABLE IV
VOLUMETRIC CHANGE OF EROSION, DEPOSITION, AND NET CHANGE ON THE VALIDATION AREA OF SC_QG

Study site	Method	Erosion (m^3)	Deposition (m^3)	Net change (m^3)
Selected slope of SC_QG	Raw DoD	-86.65	28.33	-58.32
	Raw EST	-6.28	1.86	-4.42
	Adapted EST	-3.29	0.65	-2.64
	M3C2	-1.91	1.01	-0.9

The comparison of DEMs with the limited amount of GCPs and statistical indicators such as RMSE and STD are the most commonly employed strategy for assessing DEM random errors [64]. However, this initiative failed to reveal the spatial heterogeneity of random errors, especially in complex terrain environments where GCPs are not available to surveyors [12]. Compared to simple scenarios (flat terrain with lack of vegetation), researchers preferred to focus on areas with complex topographic environments (steep terrain with vegetated areas), which tended to be hotspots for geomorphic evolution (e.g., debris flows, landslides, and gully erosion). In this study, we evaluated the random errors of fused point cloud-derived DEMs on steep slopes by global statistical analysis with TLS scanned point cloud data of the steep slopes as a benchmark. Compared to the limited number of GCPs, each of the scanning points can be regarded as an individual measurement, effectively avoiding potentially erroneous measurements results. In addition, we also expressed the spatial patterns of DEM random errors and quantified the relationships between DEM random errors and influential factors (including slope, terrain roughness, point cloud density, and vegetation cover). The million number of samples (number of cells or points) motivated a more reliable empirical relationship than previous studies (using limited GCPs). Spatial variability in DEM random errors and these empirical relationships (e.g., between slope and DEM error) offered a vital fundament for modeling the spatial pattern of random errors [42]. To the best of our knowledge, few previous studies have provided such a comprehensive and detailed analysis of DEM errors. Our study thus provided a better understanding of the errors in DEMs and provide insights for the development of DoD uncertainty estimation methods in GCD.

B. Adapted DoD Uncertainty Estimation Method and Volumetric Changes

Following the in-depth understanding of DEM errors, the adapted DoD uncertainty estimation method considered spatial pattern of random errors in a cell by cell basis by the use of a brand-new five inputs FIS model requiring five inputs, including two original variables (slope and point density) and three new parameters (topography roughness, vegetation cover, and interpolation error). Due to the flexible adjustment of the FIS inputs parameters, many studies constructed their respective FIS models to derive spatial patterns of DEM errors based on local study conditions [38], [40], [65] or employing different instruments [66], [67]. Several studies have assessed differences in deriving spatial patterns of DEM errors for different numbers of FIS input parameters [42]. However, the derived DEM error from those studies were commonly determined empirically by experts (Wheaton et al. [33]), which was not beneficial for the transferability of the method. In contrast, the adapted DoD method overcame this drawback. The FIS model membership functions setting and magnitude of random errors were derived with reference to the quantitative relationship between influencing factors and the DEM errors, which significantly improved the reliability and transferability of the improved method.

Furthermore, the proposed method also accounted for the magnitude of systematic bias on the derived DoD uncertainty. The framework of EST method did not consider the effect of systematic errors and simply assumed that the data were unbiased between different periods [43]. However, recent studies have demonstrated the widespread existence of systematic bias between point cloud dataset collected in different periods, due to the differences in laser equipment and data acquisition settings [45], [68]. Even though the same equipment and flight design were implemented in this study, there were still small systematic biases in the datasets acquired in different times. The slight residual systematic bias after co-registration process may have little effect on the elevation change results, while this procedure was necessary to perform a robust assessment in the co-registration process. In any case, the effort of describing the residual errors should be considered in the signal of the changes, particularly with the extension of study area [34]. A simple error estimation was sufficient for the studies with a large magnitude of changes (e.g., debris flows) [69]. However, a cautious errors analysis was essential for such relatively small change signals (slope erosion) [48].

The adapted DoD uncertainty estimation method overestimated the erosion volume and underestimated the deposition volume for the selected slope compared to the TLS results employing M3C2 algorithm. There were a number of reasons for this. The first one is the difference in the data source. During RAV flights, the scanner could only capture the topographic information from a top-down perspective, and even adopted the intersecting-parallel flight route, there was still a lack of terrain details on the selected slope, especially in the steep inward concave areas. In contrast, the TLS instrument performed separately at different locations beside the slope and thus effectively captured the detailed topographic information. Furthermore, the point cloud acquired by ULS was limited by the precision of the equipment, resulting in the DoD method failing to separate the DoD uncertainty and the slight elevation change. In particular, it was rather difficult to capture the changes induced by erosion/deposition when the magnitude of the true change was on the same order of the DoD uncertainty (e.g., at the base of the slope in the SC_QG) [34]. The higher precision TLS point cloud data achieved more changes than the adapted method using TLS

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data, since it could easily capture such subtle topographic changes. Second, a considerable difference in the principle of DoD method and M3C2 method. The DoD is a DEM-based method, which requires the interpolation of the point dataset that thus induces in extra interpolation error in the derived DEMs [18]. Even though the proposed method estimated the DEM error taking the effect of interpolation errors into account, any method is subject to imperfections in estimating the errors [34], and a small amount of interpolation error may still be retained in the true change, especially in areas of steep terrain (presenting higher interpolation error). The M3C2 method, estimates the topographic changes directly based on the 3-D points, skipping the DEM generation procedure [18]. The difference may also results in an overestimation of the erosion by the proposed method as compared to the M3C2 algorithm. Finally, we additionally compared the difference of elevation change between the original EST method and the adapted method at two study sites. Results showed that the adapted method was more reliable in capturing erosional and depositional changes. This demonstrated the robustness and reliability of the adapted method.

C. Implications and Limitations

Our work was both scientifically relevant and practically applicable, especially for the implementation of RAV remote sensing for geomorphic change studies in complex topography environments. We provided a comprehensive in-depth understanding of cell-by-cell DEM errors, which could be used for the development of errors mitigation strategies and data acquisition designs. Subsequently, the assessment of DEM errors was employed to develop a new DoD uncertainty estimation method and quantify geomorphic changes in the topographically complex environment. The proposed method can also be directly extended to monitor the volume of erosion and deposition in other catchments and study different soil erosion processes (such as gully erosion and mass movement). This might be an important advancement in catchment erosion monitoring since randomly occurring gully erosion is typically difficult to be monitored, whereas it is the main source of sediment production in the catchment. Furthermore, the proposed methods can be extended for other domains of natural hazards monitoring, such as landslides, debris-flows, volcano activity, and coastal erosion [21]. However, it is undeniable that the AEST method requires to assess the accuracy of each DEMs, which implies the necessity of field measurements with sufficiently ground checkpoints. Besides, the adapted method has not been practiced to multitemporal geomorphic change monitoring. In the future work, we will extend the method to multitemporal analyses, demonstrating the capacity for long-term GCD.

V. Conclusion

In this study, we comprehensively analyzed the accuracy (systematic bias) and precision (spatial pattern of random error) of DEMs derived by RAV point clouds for the topographically complex area in the hilly and gully loess plateau. We also produced the empirical relationship between impact factors (slope, point density, vegetation cover, and roughness) and random errors. On this basis, an AEST method was developed to derive the DoD uncertainty and employed to derive the spatial pattern of erosional and depositional changes of different geomorphic units in two study sites (SC_QG and SC_TLG). The reliability of the AEST method was verified by the TLS-derived results in the SC_QG and by the erosion pins result in the SC_TLG, respectively.

Results showed that the systematic bias was remarkably reduced after the co-registration procedure for SC_QG and SC_TLG, respectively. The random errors presented an approximate Gaussian distribution, and the high values were located in areas with steep topography for the SC_QG. Significant statistical relationships ($p < 0.01$) were found between random errors and influencing factors, including slope, point density, vegetation cover, and roughness. The spatial pattern of volumetric changes derived by the adapted method were considerably different from raw DoD method. Erosional volume were estimated to be -252.29 and -981.07 m^3 for the SC_QG and SC_TLG, while the corresponding depositional volume were 30.57 and 1594.32 m^3, respectively.

Spatial pattern of geomorphic changes derived by the AEST method generally was comparable to that of the M3C2-derived. Besides, the AEST method showed quite low absolute error (0.034 m) as compared to erosion pin measurements in the SC_TLG. The AEST method could capture the slight depositional changes in both study sites and performed better than the original EST method, which have potential for broader-scale soil erosion monitoring.

In summary, the AEST method can more accurately monitor the geomorphic change, especially capturing slight change in flat terrain. This method can also be directly extended and applied to other source point cloud data (e.g., mobile scanning point clouds), but the related accuracy should be revalidated. In addition, the adapted method has not yet carried out multitemporal surface process studies, which will be *a*-direction of future work. Finally, the improved EST method can be employed not only to monitor soil erosion processes but also to other surface change processes.

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